Which interchange should the Airport Rail Link plan its first bus bridge for?

1 The decision

The Airport Rail Link (ARL) operations manager at Asia Era One must decide where to stage the first standby bus bridge for an interchange closure, with a plan tested by 31 Jan 2027. The answer moves the buses: Phaya Thai (busiest) keeps them downtown, while Hua Mak sends them to the airport end.

2 Why a network

Ranking stations by riders is easy in Excel, but it cannot show who *else* loses every rail route when one station closes. A **station graph** answers that reachability question across 193 stations.

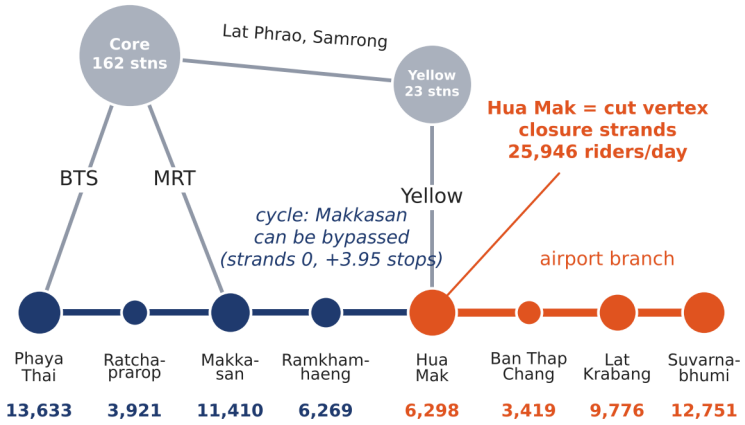
3 Data

Sources	(a) Dept. of Rail Transport: daily ARL entries and exits per station (drt2566_02). (b) CC0 GitHub network (125 stations, 2023) + Yellow, Pink, Red lines from Wikipedia.
Permission	(a) Open Data Common; robots.txt blocks bots, so downloaded by hand, not scraped. (b) CC0 / CC BY-SA. No personal data.
Period	1 Jan 2022 – 31 Mar 2026: 12,408 station-days. Weights: Apr 2025 – Mar 2026 (67,476/day).
Sampling bias	Station counts exist only for the ARL, so the 185 other destinations weigh equally, and counts are not origin-destination trips. Line counts checked against ExploreMetro.

4 Graph construction

Node / edge	A station (193). An edge asserts adjacent stops on one line (186) or a transfer walkway (17).
Metric	Remove an interchange. Stranded = ARL riders × share of non-ARL destinations no longer reachable. ARL-to-ARL trips (the seed) are excluded.
Weighting	Real ARL riders, **not** all-pairs. All-pairs ranks Wat Phra Sri Mahathat 1st and Hua Mak only 6th: it measures the whole city, not airport riders.
Pruning	None: every edge is physical track.
Result / structure	193 nodes, 203 edges, 11 cycles. **Core–periphery built from chains**, with 58 **bridge** nodes (cut vertices): Hua Mak is one; Makkasan sits on a cycle.
Robustness	Hua Mak first in all 8 runs (4 years × transfer cost 1 or 3 stops) and with equal station weights.

5 The graph



the quietest interchange is the one the airport branch hangs on

Number = ARL riders/day. Grey nodes group the other 185 stations (analysis used all 193). City rank = all-pairs rank of 18.

Interchange	Riders	Stranded	City rank
Hua Mak	6,298	**25,946**	6th
Phaya Thai	13,633	0	18th
Makkasan	11,410	0	16th
Wat Phra Sri M.	off-line	7,743	1st

6 Findings

- **The busiest interchange is not the one to protect.** Phaya Thai has the most riders (13,633/day), but closing it strands nobody else; Makkasan's riders also reroute (0 stranded, +3.95 stops).
- **Hua Mak is the cut vertex.** With only 6,298 riders/day, its closure strands **25,946/day (38.5%)** at Suvarnabhumi, Lat Krabang and Ban Thap Chang. On the 2022 network (pre-Yellow) Makkasan held that role (24,348).
- **Robustness.** First in 8 of 8 runs (37.6–38.5% stranded) and with equal weights (37.5%). It fails only if closures are station-only (see §8).

7 Recommended actions

- **By 31 Jan 2027**, the ARL operations manager table-top tests a Ban Thap Chang–Ramkhamhaeng bus bridge staged at Hua Mak, sized for ~2,600 riders/peak hour (assumed 10% of 25,946).
- **By 28 Feb 2027**, replace any pre-2023 Makkasan-first plan with staff and announcements rerouting via the Yellow Line; Phaya Thai gets signage to the MRT, not buses.
- Re-run when each new line opens (Orange Line next). Switch the plan if Hua Mak stops being a cut vertex.

8 What would change my mind

- Node removal assumes the track through the station is blocked (flood, power, signal). If most closures are station-only and trains run through, nobody beyond the station is stranded and Phaya Thai's 13,633 riders make it the priority. A year of incident logs split by type would settle it before buses are contracted.